Most Common Kernel Functions

Linear Kernel

$$K(a, b) = a^T b \qquad \text{No Mapping, } \varphi(x) = x$$

Quadratic Kernel

$$K(a, b) = [a^T b]^2 \quad \text{or} \quad K(a, b) = [a^T b + 1]^2$$

Polynomial Kernel (of degree "d")

$$K(a, b) = [a^T b]^d \quad \text{or} \quad K(a, b) = [a^T b + 1]^d$$

Radial Basis Function (RBF) Kernel

$$K(a, b) = [e^{-\gamma \| a - b \|^2}]$$

γ is called RBF Kernel bandwidth